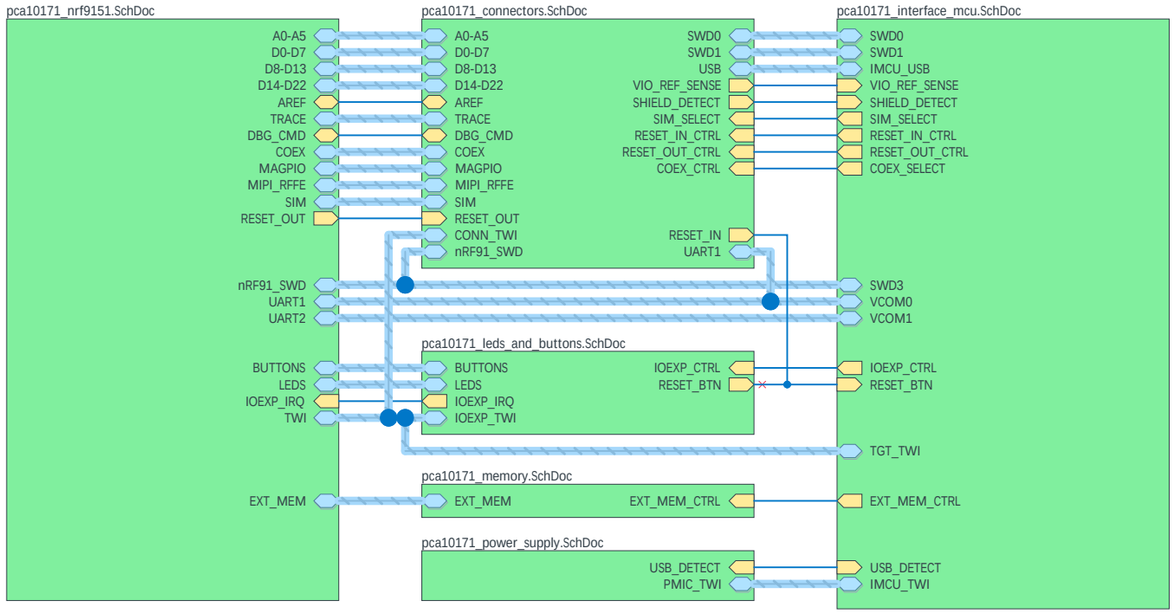
Nordic Semiconductor ASA

nRF9151 Development Kit (PCA10171)

Sheet 1: Top Level
Sheet 2: nRF9151
Sheet 3: Interface MCU
Sheet 4: LEDs and Buttons
Sheet 5: IO Expander
Sheet 6: Memory
Sheet 7: Connectors
Sheet 8: Power Supply
Sheet 9: Analog Switch 1
Sheet 10: Analog Switch 2



× The No ERC object is a design directive.
This directive is placed on a node in the circuit to suppress harmless warnings and/or error violation conditions that are detected when the schematic project is compiled.

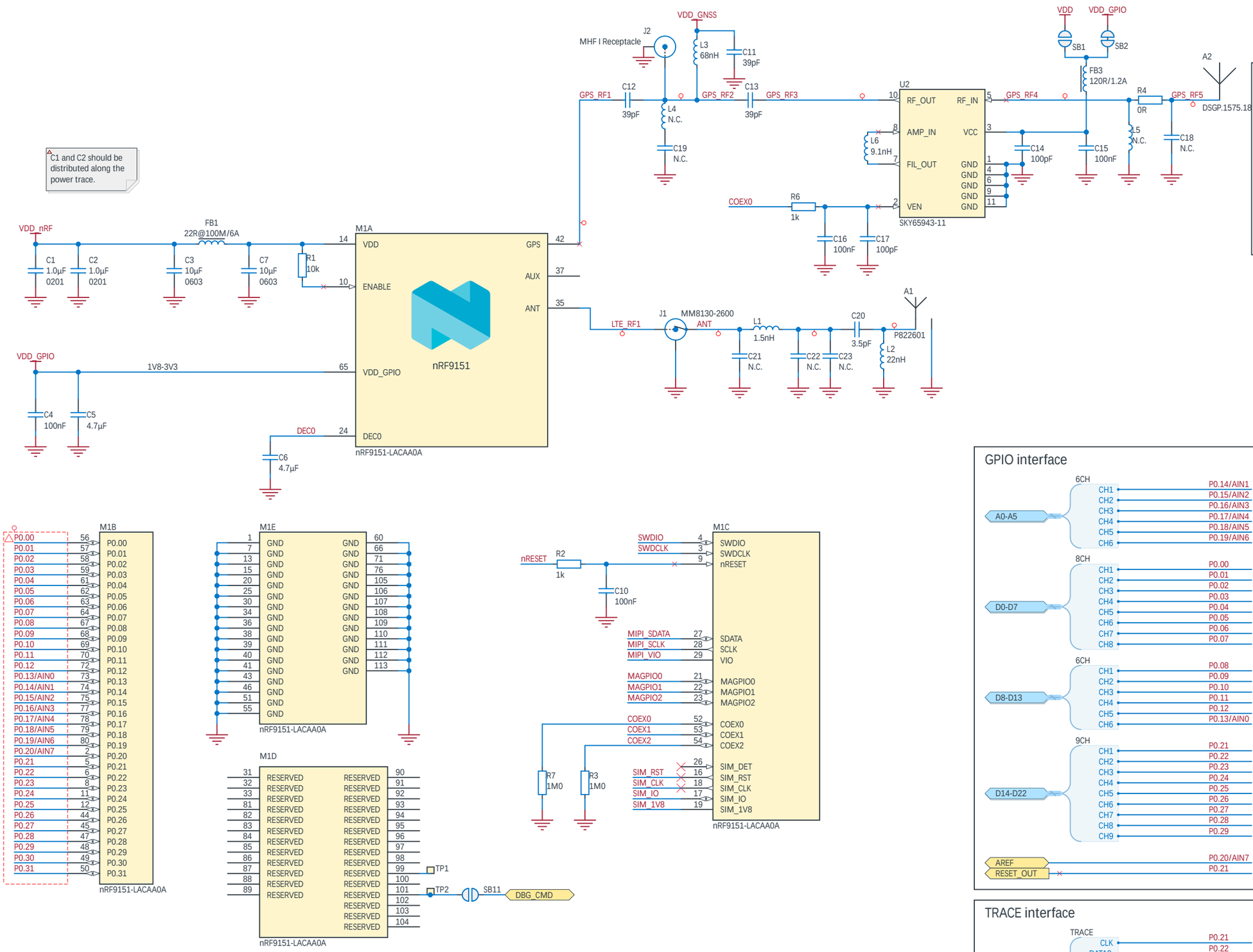
SEGGER
It simply works!

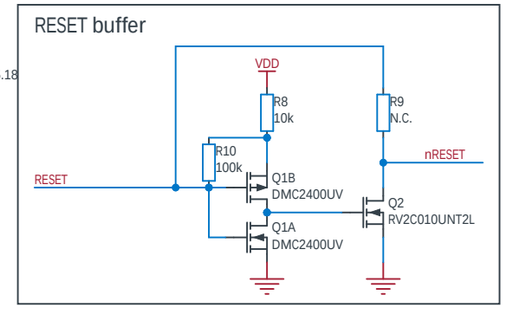
Nordic Logo

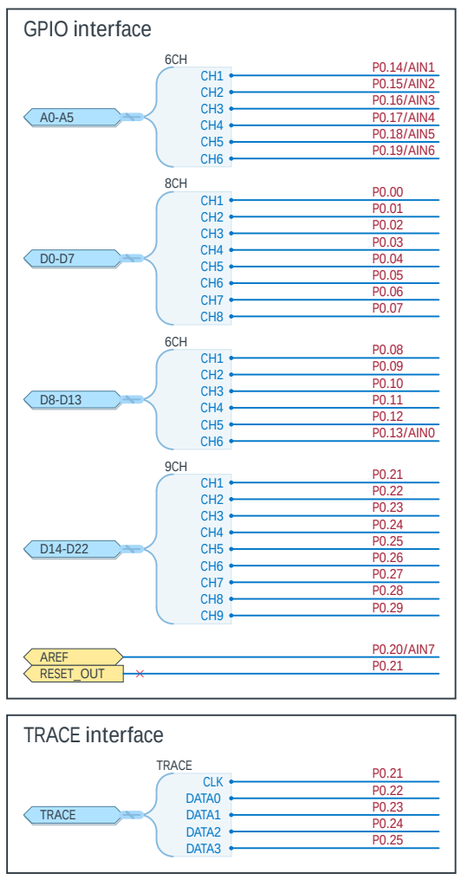
PCA10171
1.0.0
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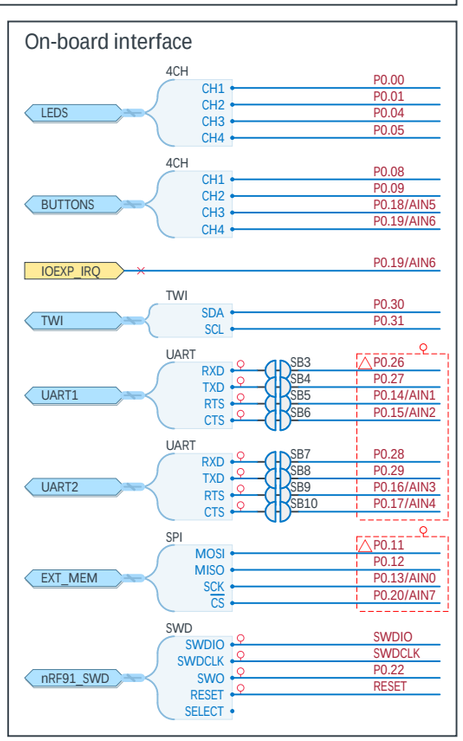
Title: nRF9151 Development Kit - Top Level		NORDIC SEMICONDUCTOR
Size: A3	PCB Assembly Number: PCA10171	Revision: 1.0.0
Date: 27/03/2024		Sheet 1 of 10
File: pca10171_top_level.SchDoc		Drawn By: KOD1
Classification: PUBLIC		

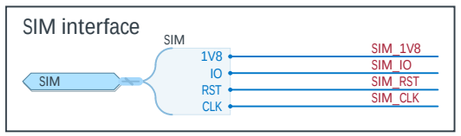
C1 and C2 should be distributed along the power trace.

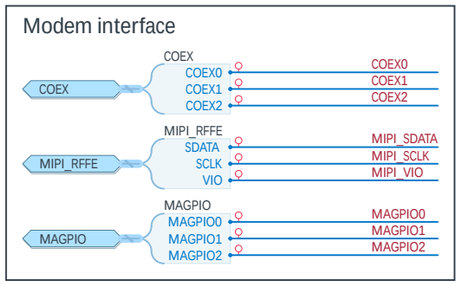












Title: nRF9151 Development Kit - nRF9151		
Size: A3	PCB Assembly Number: PCA10171	Revision: 1.0.0
Date: 29/04/2024		Sheet 2 of 10
File: pca10171_nrf9151.SchDoc		Drawn By: KOD1
Classification: PUBLIC		

DEBUGGER

Parameter Set

U3A — nRF5340-QKAA

VDD_IMCU: C28 1.0µF, C29 1.0µF, C30 2.2nF, C35 1.0µF, C36 1.0µF, C24 100nF, C25 100nF, C26 100nF, C27 100nF, C31 100nF, C32 100nF, C33 100nF, C34 100nF

DECR: A27 DECR, G31 DECRF

B8 VDD, B10 DCCD

A15 DECD

L1 VDD

J1 DCCH, E1 VDDH

VDD: A19, A21, B28, E31, N31, AL25, AL11, AL7, AG1, U1, C1, AJ31, AL17, AL3, AC1

XC2: B30 — X1 32MHz — XC1: C31

ANT: L31

RESET: AC31 — IMCU_RESET

SWDCLK: W31 — IMCU_SWDCLK

SWDIO: AA31 — IMCU_SWDIO

D-: B4 — D_N

D+: B2 — D_P

VBUS_IMCU: R21 2R2

A5 VBUS, B6 DECUSB, A23 DECN, A13 DECA

C37 4.7µF, C38 4.7µF, C39 100nF, C40 1.0µF

U3B — nRF5340-QKAA

SWD3_IO	N1	P0.00/XL1
SWD3_CLK	R1	P0.01/XL2
SWD3_SWO	W1	P0.02/NFC1
SWD3_RESET	AA1	P0.03/NFC2
VIO_REF_SENSE	V2	P0.04/AIN0
USB_DETECT	Y2	P0.05/AIN1
SWD_CTRL	AB2	P0.06/AIN2
RESET_IN_CTRL	AD2	P0.07/AIN3
RESET_OUT_CTRL	AH2	P0.08/TRACEDATA[3]
TGT_TWI_CTRL	AJ1	P0.09/TRACEDATA[2]
SWD2_SWO	AK2	P0.10/TRACEDATA[1]
SWD2_RESET	AK4	P0.11/TRACEDATA[0]
SWD2_SEL	AK6	P0.12/TRACECLK
VCOM0_RTS	AL5	P0.13/QSPI[0]
SIM_SELECT	AK8	P0.14/QSPI[1]
TGT_SDA	AK10	P0.15/QSPI[2]
VCOM0_CTS	AL9	P0.16/QSPI[3]
TGT_SCL	AK12	P0.17/QSPI_CLK
COEX_SELECT	AK14	P0.18/QSPI_CS
VCOM1_RTS	AL13	P0.19
VCOM0_HWFC	AK16	P0.20
VCOM1_CTS	AL15	P0.21
VCOM1_CTS	AK18	P0.22
VCOM1_HWFC	AK20	P0.23
RESET_BUTTON	AL27	P0.24
SWD1_CLK	AK28	P0.25/AIN4
IMCU_LED	AL29	P0.26/AIN5
SWD1_IO	AK30	P0.27/AIN6
SHIELD_DETECT	AE31	P0.28/AIN7
VCOM0_TXD	U31	P0.29
SWD0_SEL	B24	P0.30
SWD0_RESET	B22	P0.31

TP11

U3C — nRF5340-QKAA

SWD2_IO	M2	P1.00
SWD2_CLK	P2	P1.01
IMCU_SDA	AE1	P1.02/TWI
IMCU_SCL	AF2	P1.03/TWI
VCOM0_RXD	AL19	P1.04
SWD1_SEL	AK22	P1.05
VCOM1_RXD	AL21	P1.06
SWD1_RESET	AK24	P1.07
VCOM1_TXD	AL23	P1.08
SWD1_SWO	AK26	P1.09
VCOM0_CTRL	R31	P1.10
SWD0_SWO	B20	P1.11
SWD0_CLK	B18	P1.12
IOEXP_CTRL	A17	P1.13
SWD0_IO	B16	P1.14
EXT_MEM_CTRL	B14	P1.15

U3D — nRF5340-QKAA

A25	N.C.	N.C.	D2
B12	N.C.	N.C.	F2
B26	N.C.	N.C.	T2
G1	N.C.	N.C.	AG31
H2	N.C.	N.C.	A1
J31	N.C.	N.C.	A31
104	VSS_PAD	N.C.	AL1
		N.C.	AL31

UART

S1 pca10171_switch_fsa.SchDoc

VCOM0_TXD R11 150R — TP3 — VCOM0_RXD — TP4

VCOM1_TXD R12 150R — TP7 — VCOM1_RXD — TP8

VCOM0_CTRL, VCOM1_CTRL

4CH CH1 CH2 CH3 CH4 — COM — IN 1-2 — IN 3-4 — NC — NO

R13 1M0, R14 1M0

4CH CH1 CH2 CH3 CH4 — UART RXD TXD CTS RTS — VCOM0

4CH CH1 CH2 CH3 CH4 — UART RXD TXD CTS RTS — VCOM1

S2 pca10171_switch_fsa.SchDoc

VCOM0_RTS R15 150R — TP6 — VCOM0_CTS — TP5

VCOM1_RTS R16 150R — TP10 — VCOM1_CTS — TP9

VCOM0_HWFC, VCOM1_HWFC

4CH CH4 CH3 CH2 CH1 — COM — IN 1-2 — IN 3-4 — NC — NO

R85 100k, R86 100k, R17 1M0, R18 1M0

4CH CH4 CH3 CH2 CH1

4CH CH4 CH3 CH2 CH1

SWD

S3 pca10171_switch_fsa.SchDoc

SWD3_IO R19 150R, SWD3_CLK R20 150R, SWD3_SWO, SWD3_RESET R22 150R

4CH CH4 CH3 CH2 CH1 — COM — NC — IN 1-2 — IN 3-4 — NO

SWD_CTRL — R24 1M0

4CH CH4 CH3 CH2 CH1 — SWD SWDIO SWDCLK SWO RESET SELECT — SWD3

VDD — R23 N.C. — RESET_BUTTON — RESET_BTN

4CH CH4 CH3 CH2 CH1

SWD1_IO, SWD1_CLK, SWD1_SWO, SWD1_RESET, SWD1_SEL

R25 150R, R26 150R, R27 150R

SWD SWDIO SWDCLK SWO RESET SELECT — SWD1

SWD0_IO, SWD0_CLK, SWD0_SWO, SWD0_RESET, SWD0_SEL

R28 150R, R29 150R, R30 150R

SWD SWDIO SWDCLK SWO RESET SELECT — SWD0

TWI

S4 pca10171_switch_fsa.SchDoc

TGT_SDA, TGT_SCL, TGT_TWI_CTRL

4CH CH1 CH2 CH3 CH4 — COM — IN 1-2 — IN 3-4 — NC — NO

R31 1M0, R32 1M0

4CH CH1 CH2 CH3 CH4

4CH CH1 CH2 CH3 CH4 — TWI SDA SCL — TGT_TWI

DEBUGGER Programming Connector

VBUS' — TP12 — VDD_IMCU

J4 TC2050-IDC: 1 10, 2 9, 3 8, 4 7, 5 6

TP13 IMCU_SWDIO, TP15 IMCU_SWDCLK

IMCU_RESET — TP14

VDD_IMCU — R34 100k — C41 10nF

LED

VDD_3V0 — R33 120R — LED5 L0603G — Q3 RV2C010UNT2L — IMCU_LED

Board control

SIM_SELECT	SIM_SELECT
COEX_SELECT	COEX_SELECT
VIO_REF_SENSE	VIO_REF_SENSE
USB_DETECT	USB_DETECT
SHIELD_DETECT	SHIELD_DETECT
RESET_IN_CTRL	RESET_IN_CTRL
RESET_OUT_CTRL	RESET_OUT_CTRL
EXT_MEM_CTRL	EXT_MEM_CTRL
IOEXP_CTRL	IOEXP_CTRL

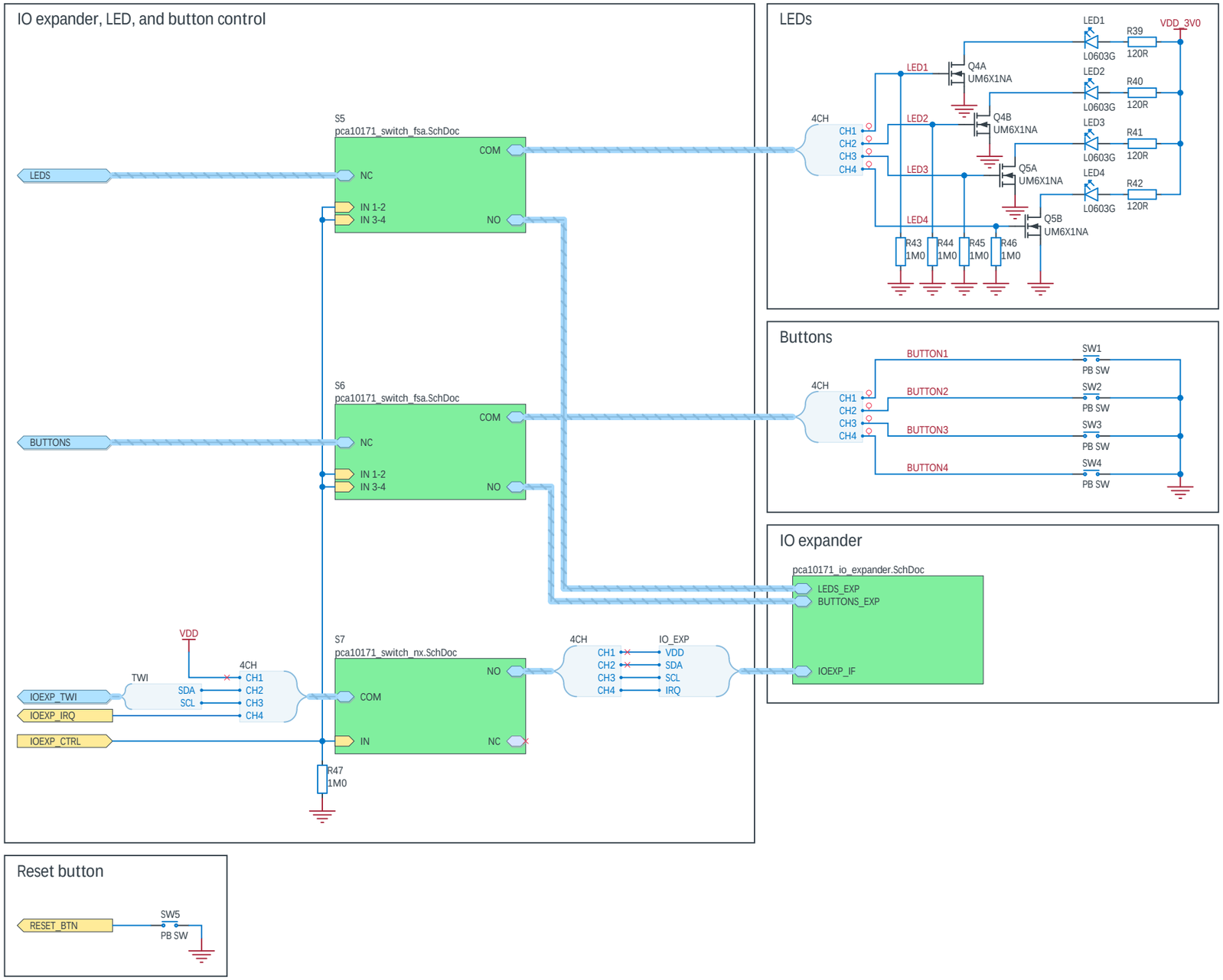
IMCU_SDA, IMCU_SCL — TWI SDA SCL — IMCU_TWI

USB

D_N, D_P — USB D-, D+ — IMCU_USB

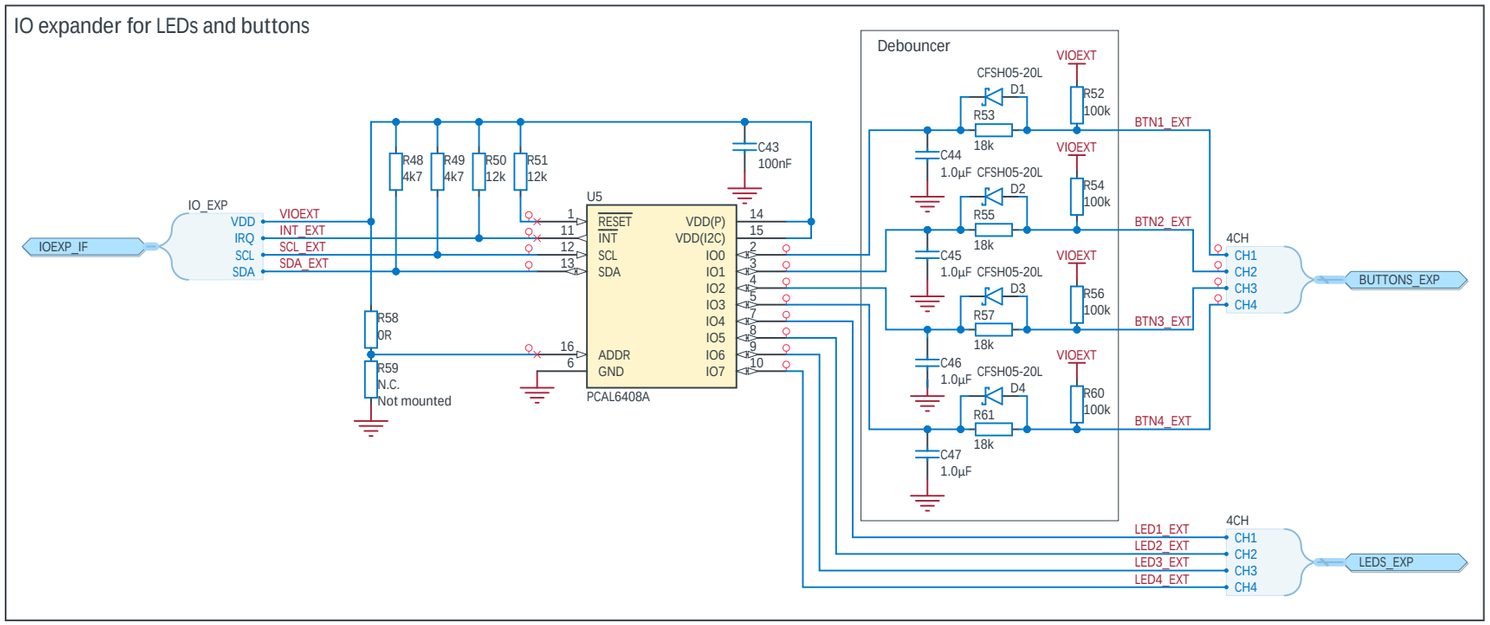
Title	nRF9151 Development Kit - Interface MCU	
Size: A3	PCB Assembly Number: PCA10171	Revision: 1.0.0
Date: 06/02/2024		Sheet 3 of 10
File: pca10171_interface_mcu.SchDoc		Drawn By: KOD1
Classification: PUBLIC		

NORDIC SEMICONDUCTOR

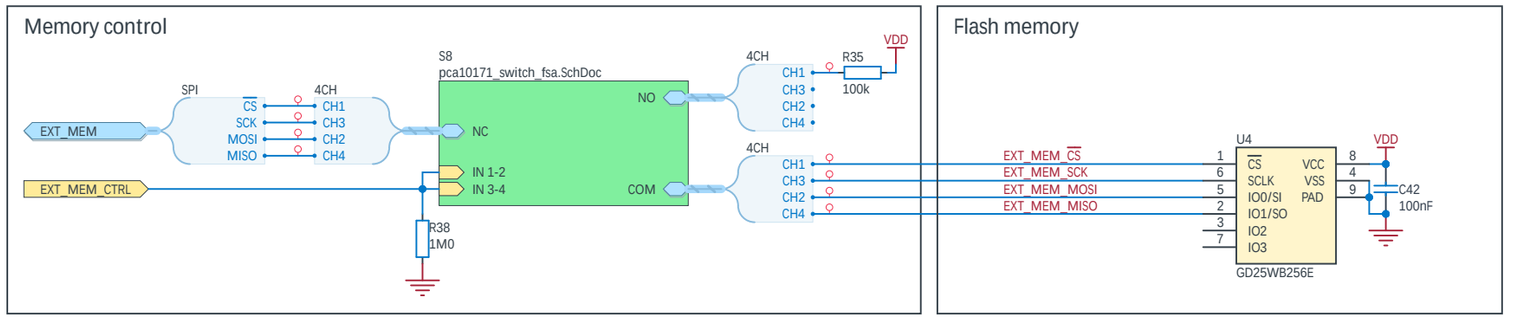


Title	nRF9151 Development Kit - LEDs and Buttons		
Size	PCB Assembly Number	Revision	
A3	PCA10171	1.0.0	
Date: 06/02/2024		Sheet 4 of 10	
File: pca10171_leds_and_buttons.SchDoc		Drawn By: KOD1	
Classification: PUBLIC			

IO expander for LEDs and buttons

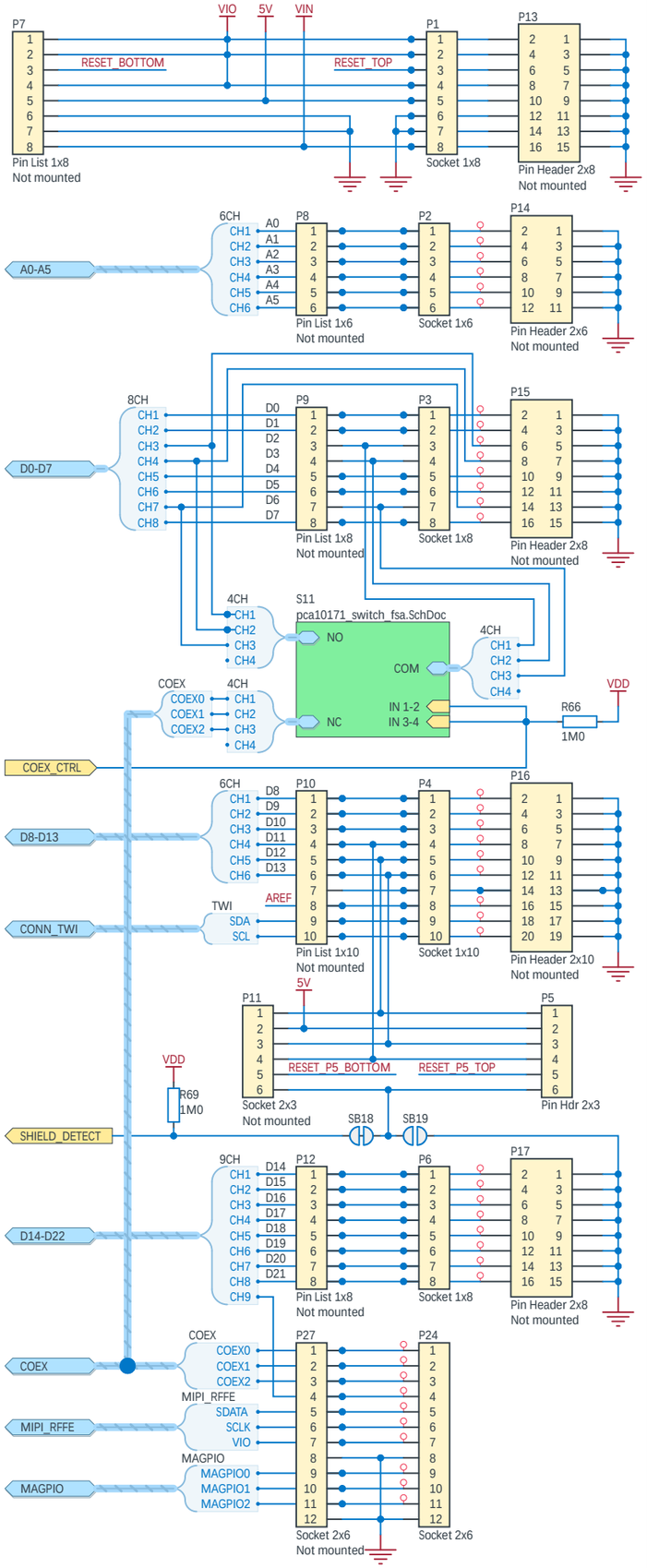


Title	nRF9151 Development Kit - IO Expander		
Size	PCB Assembly Number	Revision	
A3	PCA10171	1.0.0	
Date: 06/02/2024		Sheet 5 of 10	
File: pca10171_io_expander.SchDoc		Drawn By: KOD1	
Classification: PUBLIC			

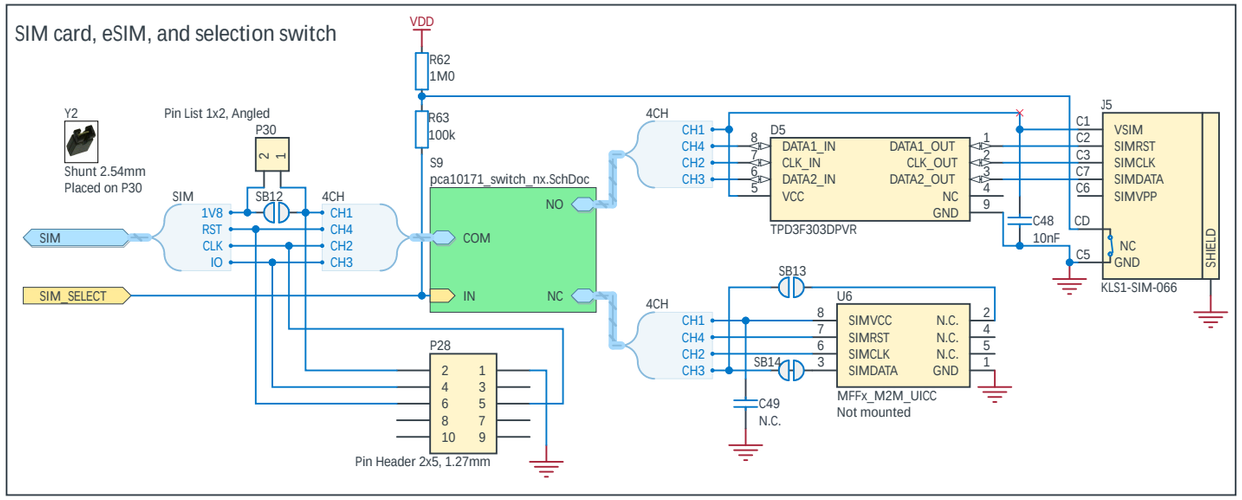


Title: nRF9151 Development Kit - Memory		
Size: A3	PCB Assembly Number: PCA10171	Revision: 1.0.0
Date: 22/12/2023		Sheet 6 of 10
File: pca10171_memory.SchDoc		Drawn By: KOD1
Classification: PUBLIC		

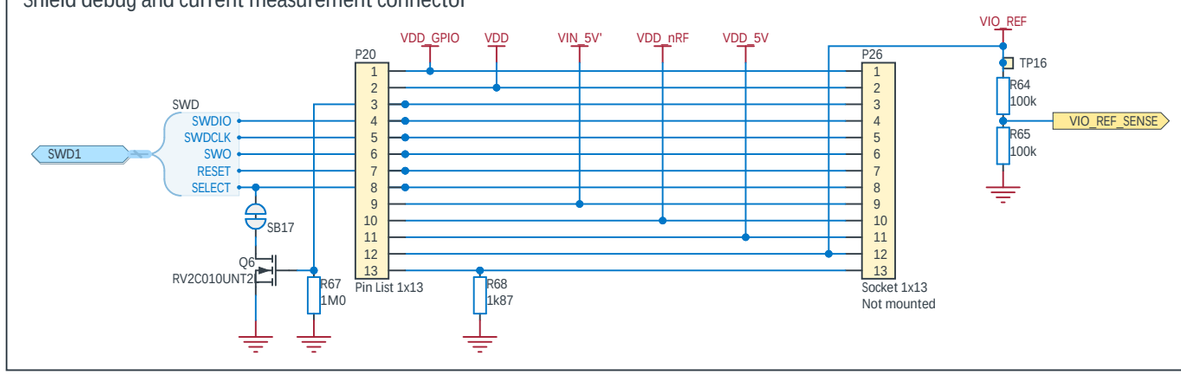
Interface connectors



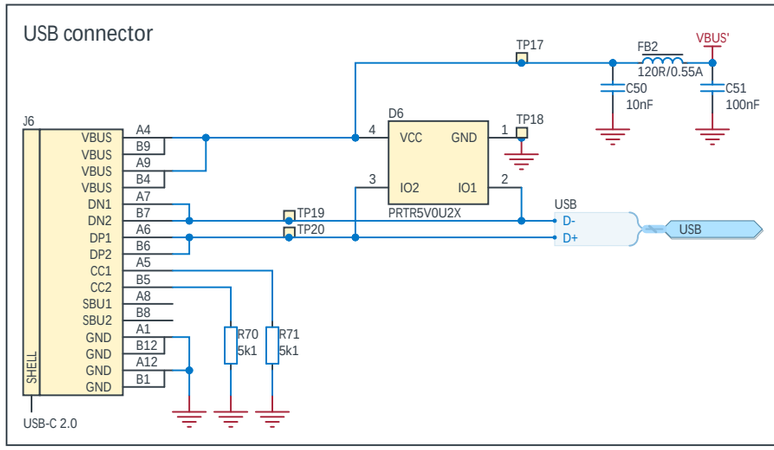
SIM card, eSIM, and selection switch



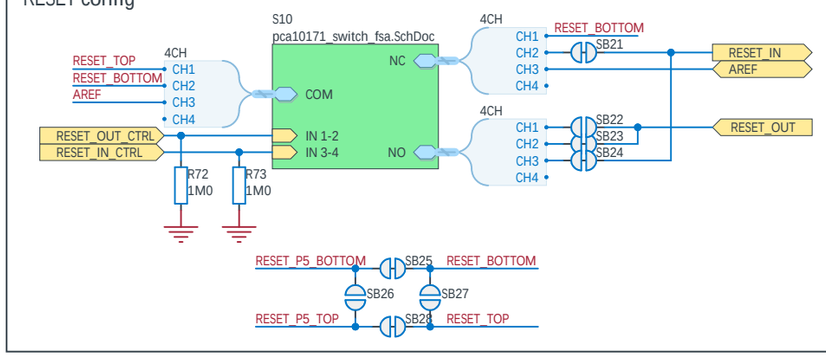
Shield debug and current measurement connector



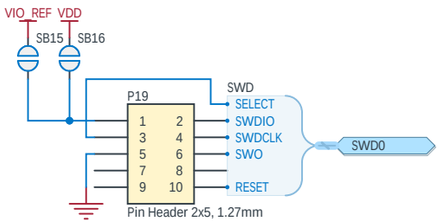
USB connector



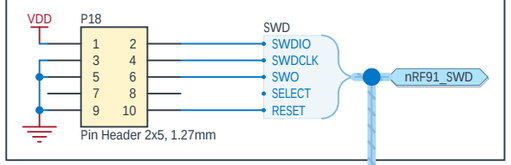
RESET config



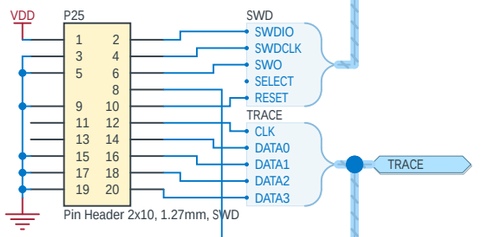
Debug OUT connector



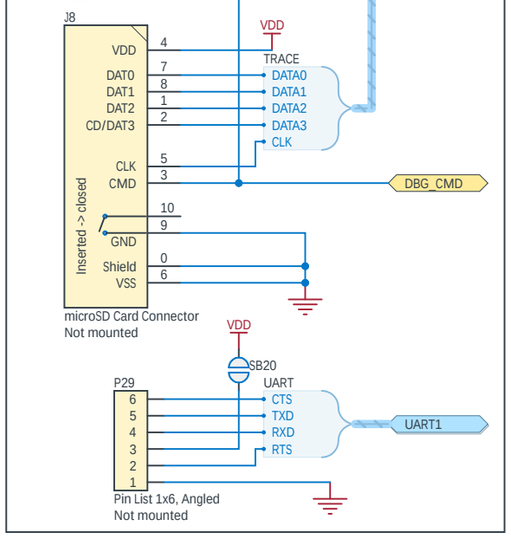
Debug IN connector



Trace connector



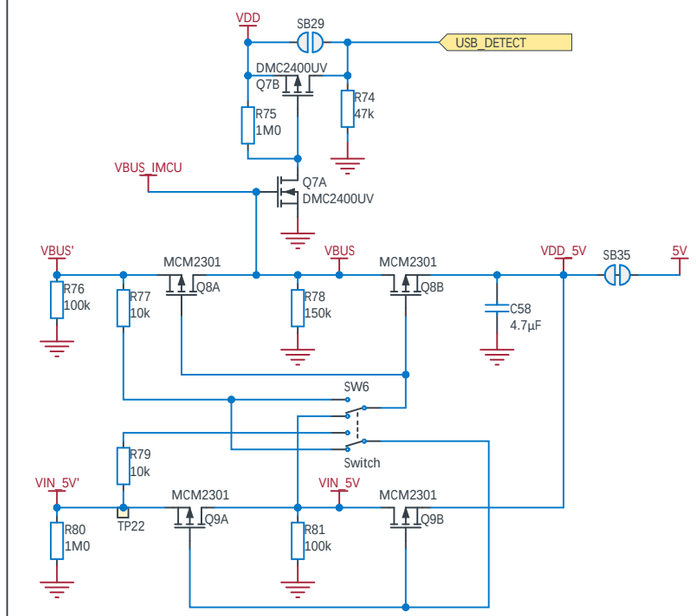
Debugging interface



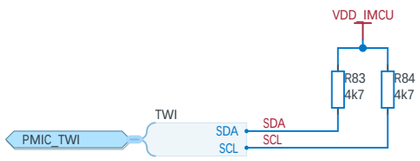
Title: nRF9151 Development Kit - Connectors		
Size: A3	PCB Assembly Number: PCA10171	Revision: 1.0.0
Date: 27/03/2024		Sheet 7 of 10
File: pca10171_connectors.SchDoc		Drawn By: KOD1
Classification: PUBLIC		

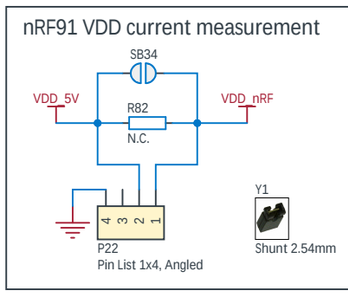
NORDIC SEMICONDUCTOR

Power input selection and power switch

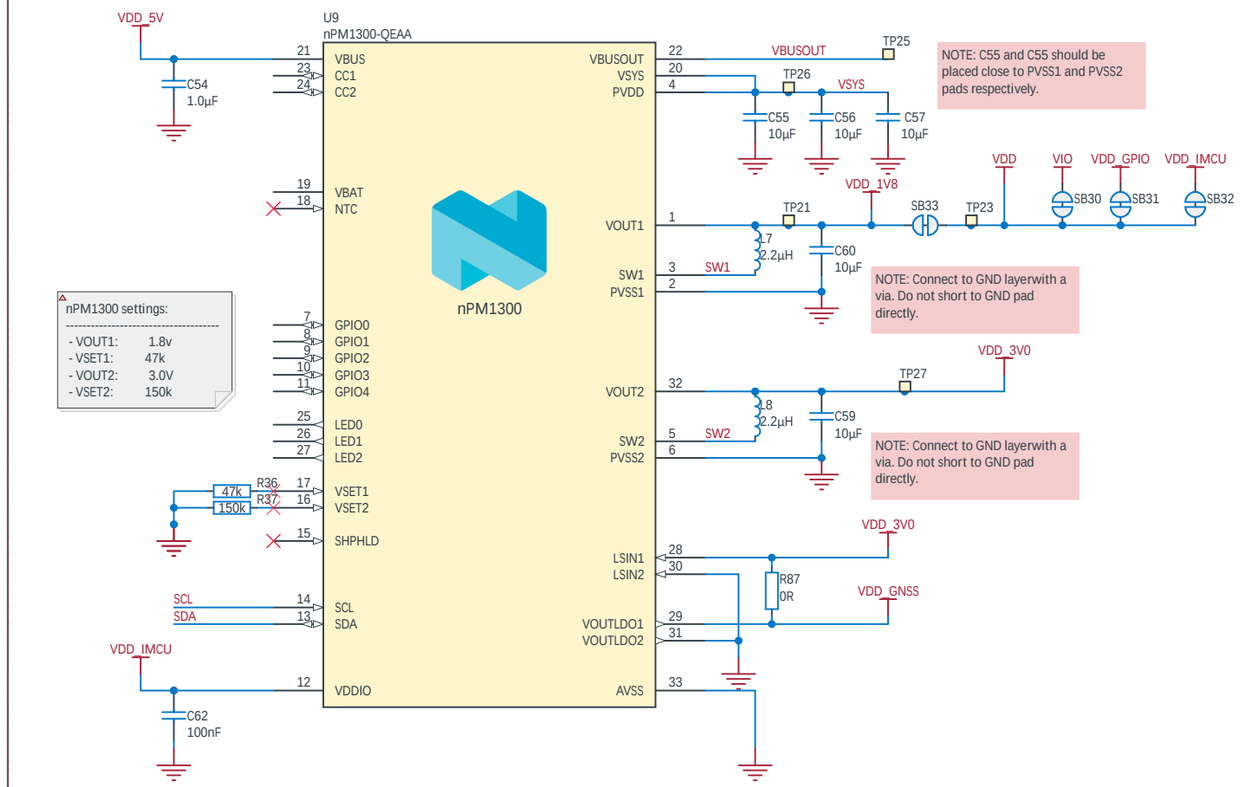


nRF91 VDD current measurement





Power managment

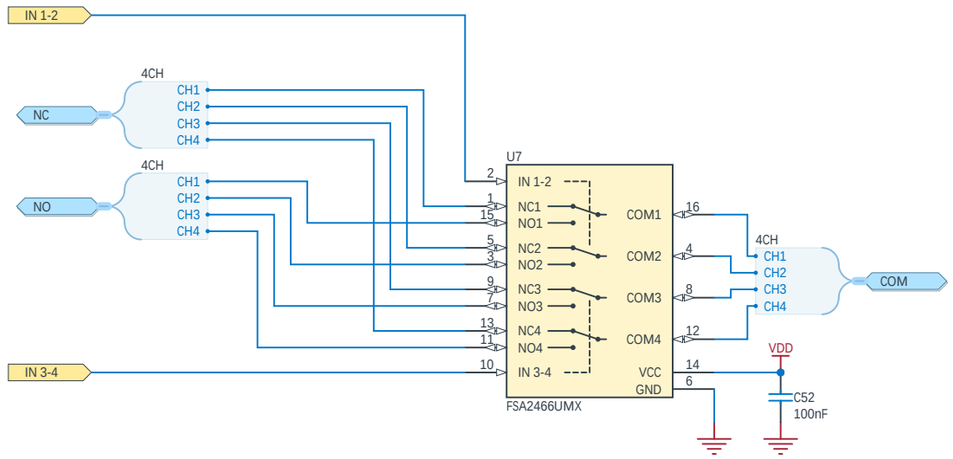


Instrument ground

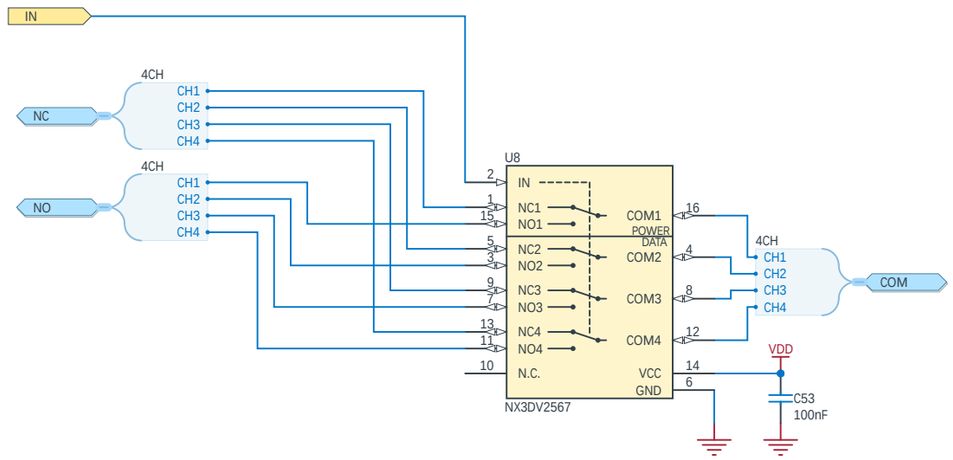


Title: nRF9151 Development Kit - Power Supply		
Size: A3	PCB Assembly Number: PCA10171	Revision: 1.0.0
Date: 07/02/2024		Sheet 8 of 10
File: pca10171_power_supply.SchDoc		Drawn By: KOD1
Classification: PUBLIC		

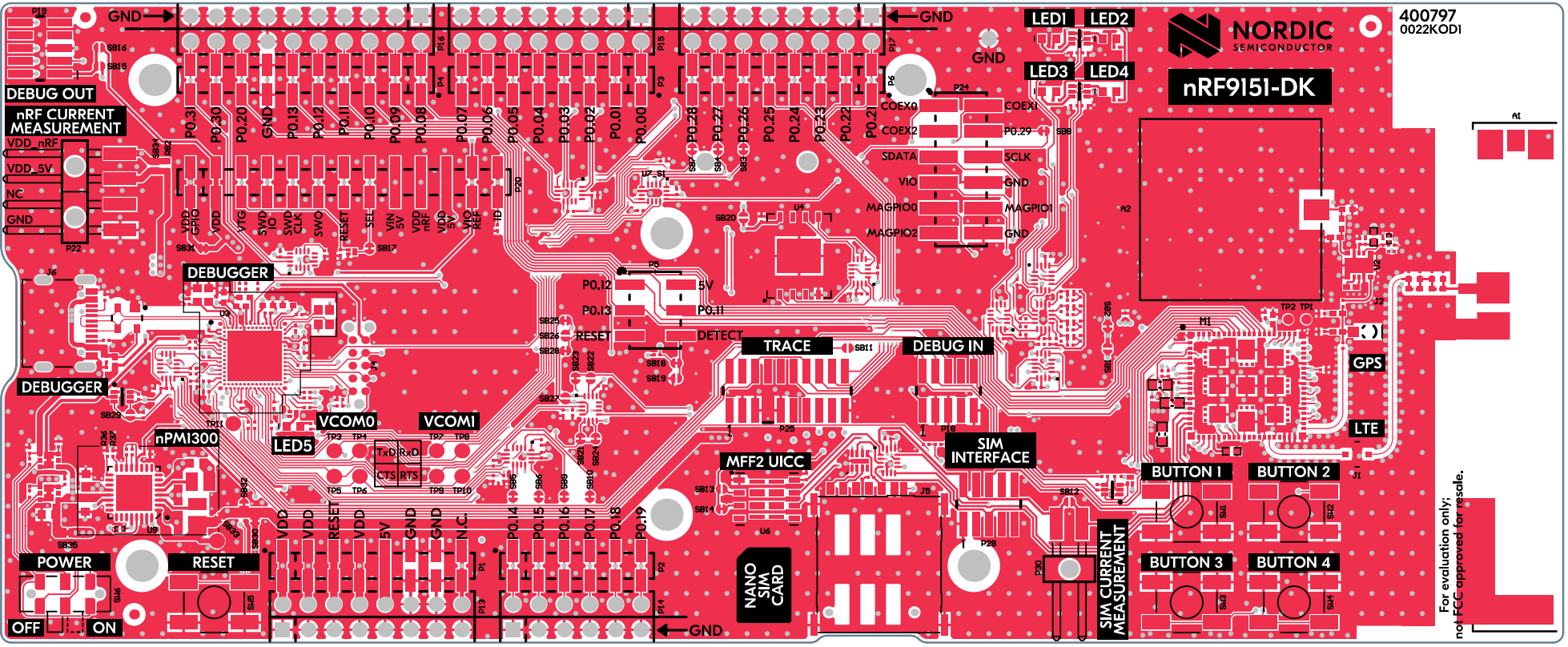
NORDIC SEMICONDUCTOR

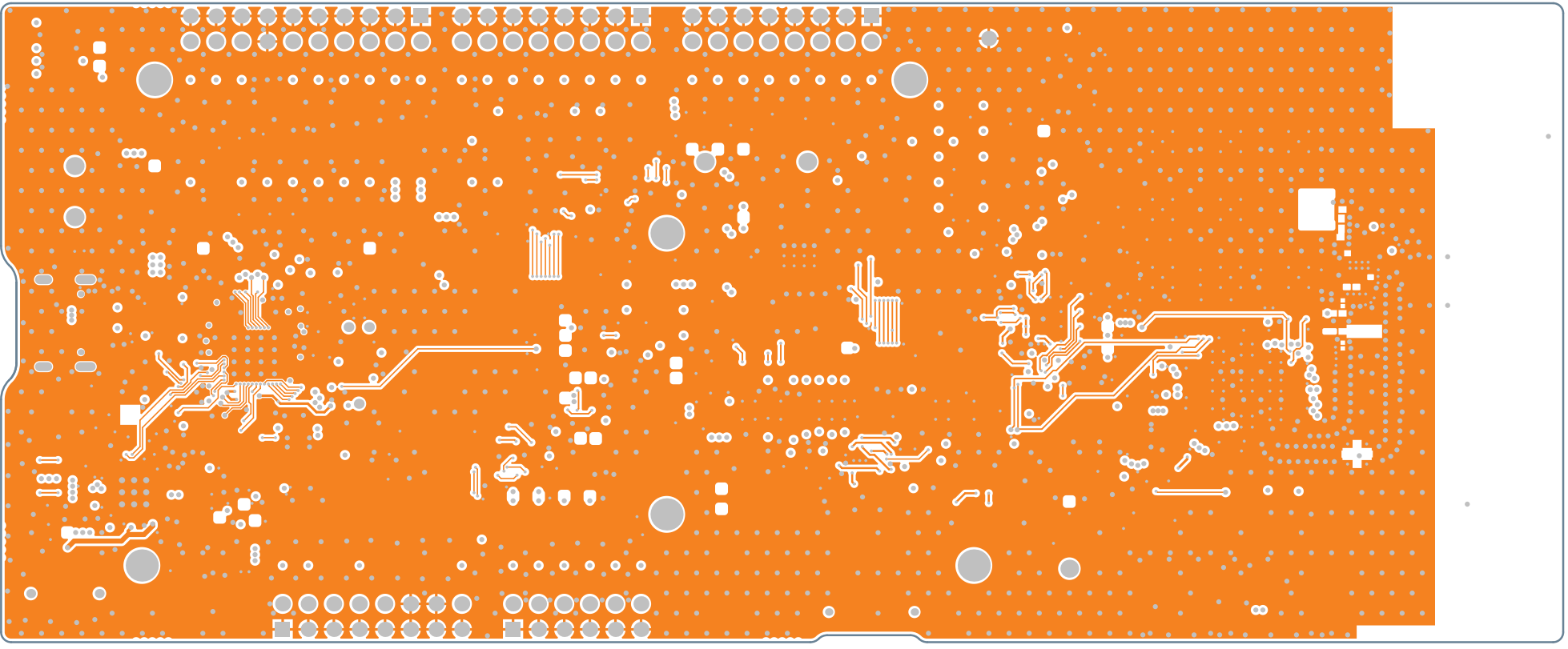


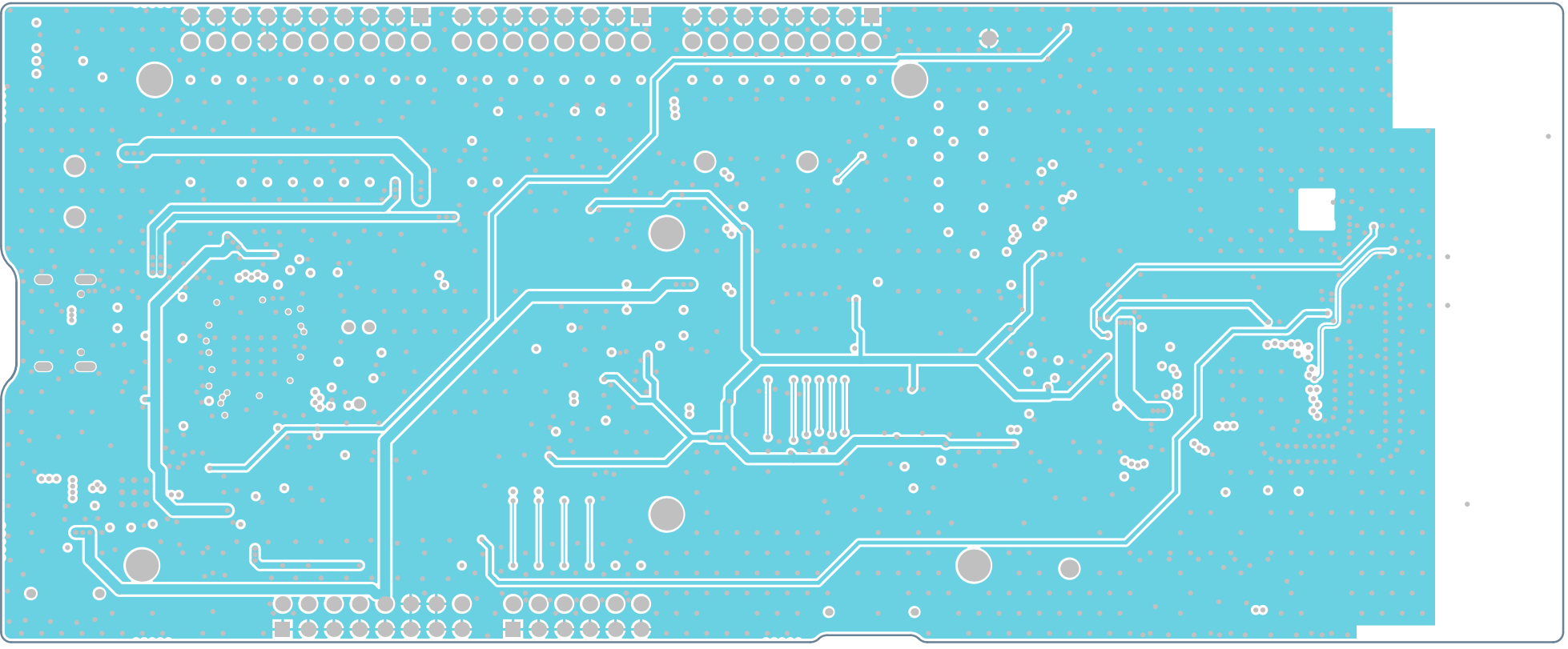
Title: nRF9151 Development Kit - Switch FSA		
Size: A3	PCB Assembly Number: PCA10171	Revision: 1.0.0
Date: 22/12/2023		Sheet 9 of 10
File: pca10171_switch_fsa.SchDoc		Drawn By: KOD1
Classification: PUBLIC		



Title		
nRF9151 Development Kit - Switch NX		
Size	PCB Assembly Number	Revision
A3	PCA10171	1.0.0
Date: 22/12/2023		Sheet 10 of 10
File: pca10171_switch_nx.SchDoc		Drawn By: KOD1
Classification: PUBLIC		


nRF9151-DK
NORDIC SEMICONDUCTOR
400797
0022KOD1
DEBUG OUT
nRF CURRENT MEASUREMENT
VDD_nRF
VDD_5V
NC
GND
DEBUGGER
DEBUGGER
nPM1300
LED5
VCOM0
VCOM1
TxD
RxD
CTS
RTS
POWER
RESET
OFF
ON
LED1
LED2
LED3
LED4
COEX0
COEX1
COEX2
P0.29
SDATA
SCLK
VIO
GND
MAGPIO0
MAGPIO1
MAGPIO2
GND
P0.12
5V
P0.13
P0.11
RESET
DETECT
TRACE
DEBUG IN
MFF2 UICC
NANO SIM CARD
SIM INTERFACE
SIM CURRENT MEASUREMENT
BUTTON 1
BUTTON 2
BUTTON 3
BUTTON 4
GPS
LTE
For evaluation only;
not FCC approved for resale.





Circuit	Function	GPIO
LED 1		P0.00
LED 2		P0.01
LED 3		P0.04
LED 4		P0.05
Button 1		P0.08
Button 2		P0.09
Button 3		P0.18
Button 4		P0.19

Button	LED
VDD/VIO	*
GND	*

Circuit	Function	GPIO
EXT MEMORY	MOSI	P0.11
	MISO	P0.12
	SCK	P0.13
	CS	P0.30

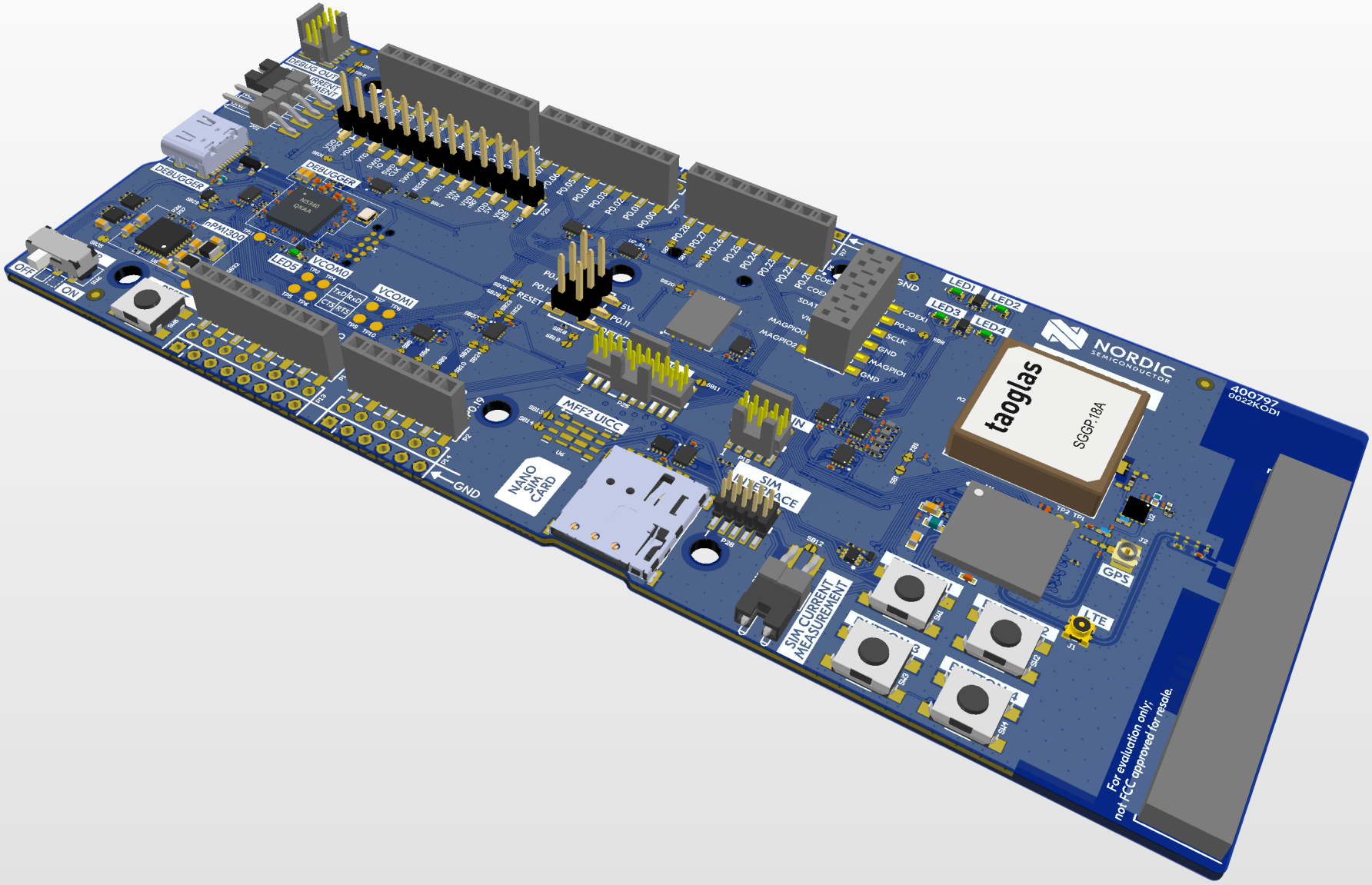
Circuit	Function	GPIO
	TXDI	P0.06
	RXDI	P0.07
	RTSI	P0.14
	CTSI	P0.15
	TXDO	P0.26
	RXDO	P0.28
	RTSO	P0.18
	CTSO	P0.17

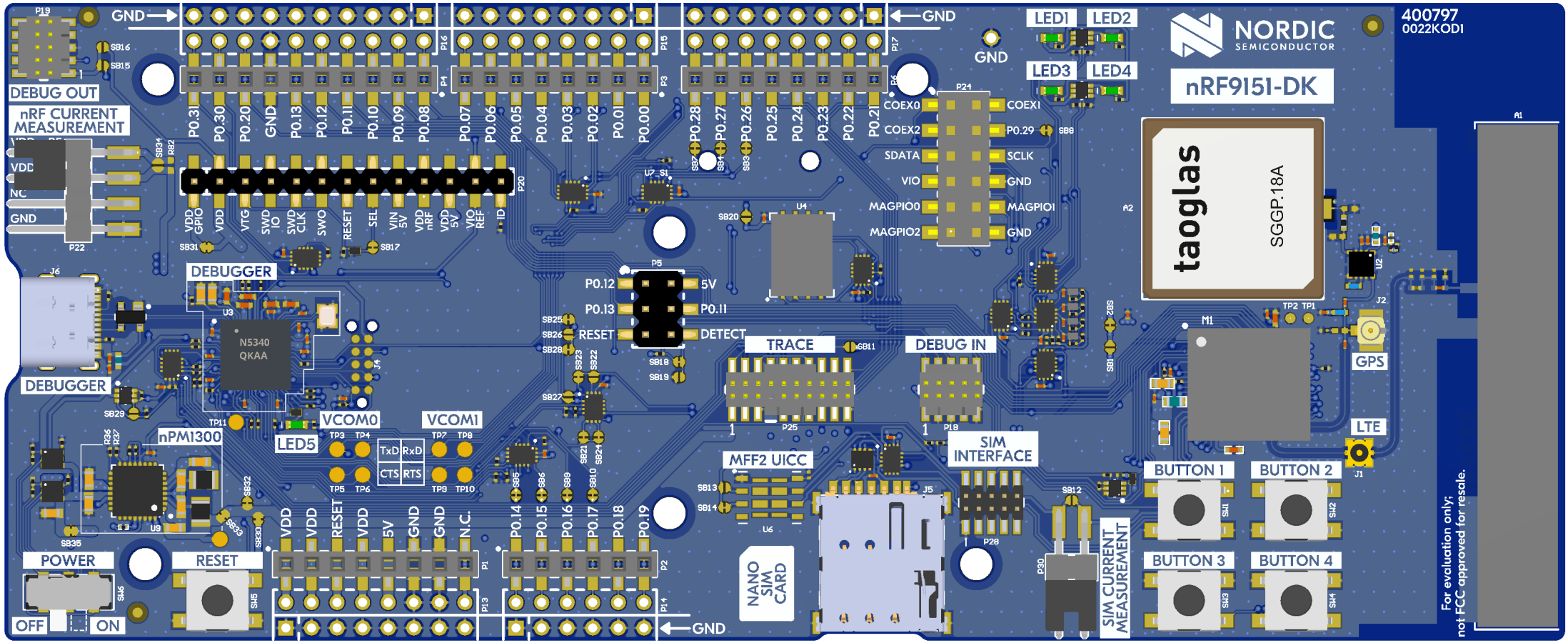
Circuit	Alt.	Default	Arduino
	P0.31	RESET	RESET
	RESET	P0.21	AREF
	P0.00	COEX3	D8
	P0.03	COEX1	D3
	P0.02	COEX0	D2

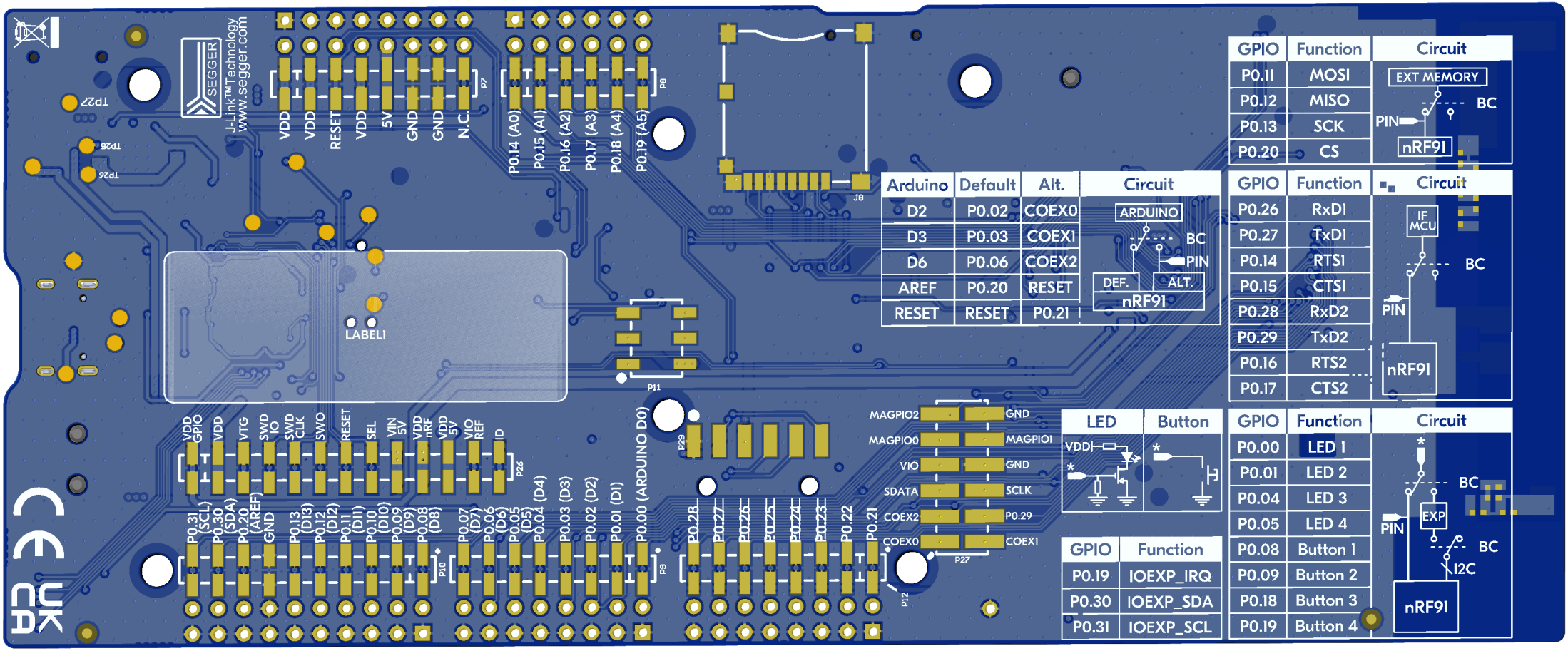
Function	GPIO
IOEXP_IRQ	P0.31
IOEXP_SDA	P0.30
IOEXP_SCL	P0.13

Function	GPIO
COEX0	P0.02
COEX1	P0.03
COEX2	P0.00
RESET	P0.21
SCLK	P0.25
SDATA	P0.24
VIO	
GND	
MAGPIO0	P0.27
MAGPIO1	P0.26
MAGPIO2	P0.23
GND	

Function	GPIO
SWDIO	P0.20
SWCLK	P0.22
GND	
VIO	
VIO	
GND	
VIO	
VIO	
GND	
VIO	
GND	
VIO	
GND


DEBUG OUT
CURRENT MEASUREMENT
DEBUGGER
nPM1300
OFF
ON
LED5
VCOM0
VCOM1
RESET
MFF2 UICC
NANO SIM CARD
SIM INTERFACE
SIM CURRENT MEASUREMENT
GND
LED1
LED2
LED3
LED4
taoglas
SGGP.18A
NORDIC SEMICONDUCTOR
GPS
LTE
400797
0022KOD1
For evaluation only;
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400797
0022KOD1
NORDIC SEMICONDUCTOR
nRF9151-DK
taoglas
SGGP.18A
LED1
LED2
LED3
LED4
GND
DEBUG OUT
nRF CURRENT MEASUREMENT
DEBUGGER
N5340 QKAA
nPM1300
LED5
VCOM0
VCOM1
TxD
RxD
CTS
RTS
TRACE
DEBUG IN
MFF2 UICC
SIM INTERFACE
NANO SIM CARD
SIM CURRENT MEASUREMENT
GPS
LTE
BUTTON 1
BUTTON 2
BUTTON 3
BUTTON 4
POWER
OFF
ON
RESET
P0.12
P0.13
RESET
5V
P0.11
DETECT
COEX0
COEX1
COEX2
P0.29
SDATA
SCLK
VIO
GND
MAGPIO0
MAGPIO1
MAGPIO2
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Arduino	Default	Alt.	Circuit
D2	P0.02	COEX0	ARDUINO, BC, PIN, DEF., ALT., nRF91
D3	P0.03	COEX1	
D6	P0.06	COEX2	
AREF	P0.20	RESET	
RESET	RESET	P0.21	

LED	Button
VDD	*
*	

GPIO	Function
P0.19	IOEXP_IRQ
P0.30	IOEXP_SDA
P0.31	IOEXP_SCL

GPIO	Function	Circuit
P0.11	MOSI	EXT MEMORY, BC, PIN, nRF91
P0.12	MISO	
P0.13	SCK	
P0.20	CS	

GPIO	Function	Circuit
P0.26	RxD1	IF MCU, BC, PIN, nRF91
P0.27	TxD1	
P0.14	RTS1	
P0.15	CTS1	
P0.28	RxD2	
P0.29	TxD2	
P0.16	RTS2	
P0.17	CTS2	

GPIO	Function	Circuit
P0.00	LED 1	*, BC, EXP, PIN, BC, I2C, nRF91
P0.01	LED 2	
P0.04	LED 3	
P0.05	LED 4	
P0.08	Button 1	
P0.09	Button 2	
P0.18	Button 3	
P0.19	Button 4	